Seat No.:_____

AHU ID No._____ Name_____ Major_____ Grade_____

装 订 线

Stony Brook Institute at Anhui University

2022 Fall Semester

Applied Calculus I Midterm Exam

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	V	VI	Total score
Score							
Grader							

I. Completions (5 items, 3 points each)

Score	

1. If $|x| \le 1$, then $\arcsin x + \arccos x =$ _____.

2. Compute the limit $\lim\limits_{n\to\infty}\left(\sqrt[n]{1}+\sqrt[n]{2}+\cdots+\sqrt[n]{10}\right)=$ _____.

3. Evaluate the limit $\lim\limits_{x\to+\infty}\sqrt{x^2+x}-x=$ _____.

4. Find the limit $\lim\limits_{x\to 0}\dfrac{\sin(\tan x)}{x}=$ _____.

5. If $x\ln(1-ax)\sim(1-\cos x)(x\to 0)$, then $a=$ _____.

II. Choices (5 items, 3 points each)

Score	

6. Supposing that sequences $\{x_n\}$ and $\{y_n\}$ satisfy $\lim\limits_{n\to\infty}x_n=A, \lim\limits_{n\to\infty}y_n=B$, and $A\ne B$. then which one is the limit of the sequence $x_1,y_1,x_2,y_2,x_3,y_3,\cdots$ ( )

A. A B. B C. $A+B$ D. it doesn't exist

7. Find the infinity as $x\to 0$ ( )

A. $\dfrac{1}{x}\cos x$ B. $x\cos x$ C. $\dfrac{1}{x}\sin x$ D. $x\sin\dfrac{1}{x}$

8. If $f(x)=\dfrac{\sqrt{1+x+x^2}-1}{\sin 2x}$, then $\lim\limits_{x\to 0}f(x)=$ ( )

A. 3 B. $\dfrac{1}{4}$ C. 1 D. $\dfrac{1}{2}$

9. If the function $f(x)=\begin{cases}\dfrac{1-\sqrt{1-x}}{1-\sqrt[3]{1-x}} & x\neq 0\\ a & x=0\end{cases}$

is continuous at $x=0$,then the constant $a=$ (B)

A. $\dfrac{1}{2}$ B. $\dfrac{3}{2}$ C. 3 D. 1

10. Let $f(x)=\begin{cases}e^{-\frac{1}{x-1}}, & x\neq 1\\ 0, & x=1\end{cases}$, then (C)

A. $f(x)$ is continuous at the point $x=1$.

B. $f(x)$ is discontinuous but left-continuous at the point $x=1$.

C. $f(x)$ is discontinuous but right-continuous at the point $x=1$.

D. $f(x)$ is neither left-continuous nor right-continuous at $x=1$.

III. Calculations (4 items, 8 points each)

Score	

11. Evaluate the limit $\lim\limits_{n\to\infty}[\dfrac{1}{1\cdot 2}+\dfrac{1}{2\cdot 3}+\cdots+\dfrac{1}{n(n+1)}]$

12. Find $\lim\limits_{x\to 1}(1-x)\tan\dfrac{\pi x}{2}$.

13. Calculate $\lim_{x\to+\infty}\left(\frac{1+x}{2+x}\right)^{\frac{1-x^2}{1-x}}$.

14. Find $\lim_{n\to\infty}\sqrt[n]{n^3+3^n}$.

IV. Analytical Judgement Problems (8 points)

Score	

15. Let $f(x)=\frac{1}{\ln|x|}$, find the discontinuous point of the function and determine the type of discontinuous point.

V. Proof problems (3 items, 8 points each)

Score	

16. Prove $\lim_{n\to\infty}\frac{10n}{n^2+1}=0$ using the "$\varepsilon-N$" definition of a limit.

17. If $a_{n+1}=2-\frac{1}{a_n}, n=1,2,\cdots, a_1>1$,using the monotonic bounded principle to prove that the limit of sequence $\{a_n\}$ exists and find the limit.

18. If $f(x)$ is continuous on interval $(a,\ b)$,and $x_i \in (a,\ b), i = 1, 2, \cdots, n$, then there exists $\xi \in (a,\ b)$, such that $f(\xi) = \frac{1}{n}\sum_{i=1}^{n} f(x_i)$.

VI. Discussion Problem (6 points)

Score	

19. Does the equation $x^4 - 15x = 2$ at least have one root in the interval [2, 3]? Please give the reason.

18. If $f(x)$ is continuous on interval $(a,\ b)$,and $x_i \in (a,\ b), i = 1, 2, \cdots, n$, then there exists $\xi \in (a,\ b)$, such that $f(\xi) = \frac{1}{n}\sum_{i=1}^{n} f(x_i)$.

VI. Discussion Problem (6 points)

Score	

19. Does the equation $x^4 - 15x = 2$ at least have one root in the interval [2, 3]? Please give the reason.